2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)
28-29 November 2024, Southeast University, Dhaka, Bangladesh

2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) | 979-8-3315-3435-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/BECITHCON64160.2024.10962766

Desmoking and Perceptual Quality Restoration of Laparoscopic Videos Using CycleGAN

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***Abstract*—Laparoscopic surgeries, valued for their minimally invasive nature, often suffer from visual obstructions caused by smoke from electrocauterization, impacting surgical precision and safety. This paper introduces a CycleGAN-based method for desmoking and restoring the perceptual quality of laparoscopic videos. Unlike traditional dehazing techniques, our approach uses inter-channel discrepancy and dark channel prior losses to effectively minimize residual smoke. Our key contributions include developing a novel GAN-based framework tailored for surgical video enhancement and introducing smoke-specific loss functions that improve visual clarity. Evaluated on the LVQ database, our model outperforms existing methods in image quality metrics and visual assessments. A user study with 25 surgeons confirmed its practical utility and potential for clinical integration, enhancing intraoperative imaging and surgical outcomes.**

***Index Terms*—dark channel, image-to-image translation, laparoscopic video, LVQ database, Generative Adversarial Network (GAN), cycleGAN, smoke removal**

I. Introduction

Laparoscopic surgery uses a miniature camera to visualize the abdominal or pelvic cavity through small incisions, with high-resolution video being crucial for successful procedures [1]. However, laparoscopic videos are often affected by distortions such as noise, defocus, motion blur, smoke, and inconsistent illumination, which, while minimally impacting image-guided systems, can reduce clarity and affect surgical precision. Traditional methods to address these issues often involve equipment replacement or troubleshooting [2]–[4].

Artifacts like smoke from laser ablation impair tissue visibility [5], leading to increased errors and prolonged surgeries. Though medical equipment can clear smoke, it is costly and impractical, prompting the need for alternative solutions. Studies [6] use computer vision algorithms for smoke mitigation, while deep learning enhances real-time surgical navigation systems [2]. Smoke removal approaches are divided into image processing, physics-based models (DCP and atmospheric scattering), and AI methods using convolutional and generative networks [7]. Recent deep learning advances enable real-time smoke removal through conditional image generation. This research applies an adversarial framework to map images between domains, improving clarity and reducing noise. The LVQ database [8] is used for training and testing, with dataset preparation outlined in Section III. This work builds on methods from [2], [9], [10] to optimize image quality.

This study proposes a method that combines embedded guide masks from inter-channel discrepancy and dark channel prior losses with an image-to-image GAN to effectively remove smoke from laparoscopic video recordings, which are often compromised during surgery. The GAN-based image processing algorithms successfully extract smoke, addressing a major challenge in laparoscopic surgery, where smoke obstructs the surgeon's view and hampers computer vision systems. A user assessment involving 25 surgeons showed that the majority favored the method due to its practicality and aesthetic appeal in surgical applications. The results highlight the technique's potential to improve surgical outcomes and support better decision-making.

II. Related Works

This section reviews smoke removal techniques categorized into classical methods, deep learning approaches, and physical-model-based techniques. Traditional methods for desmoking, often adapted from de-hazing, employ the atmospheric scattering model, represented as:

$$\mathbf{I}(p,q) = \mathbf{H}(p,q) \cdot t(p,q) + \mathbf{L} \cdot (1 - t(p,q)) \tag{1}$$

where $\mathbf{I}(p,q)$ represents the observed image, $\mathbf{H}(p,q)$ is the clear image, $\mathbf{L}$ is the global light source, and $t(p,q)$ denotes the transmission map of the medium. Bayesian and Lagrangian methods [3], [11] are employed for combined denoising and de-smoke processing. The Dark Channel Prior

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(DCP) technique, as proposed by He et al. [12] and refined through histogram equalization [13], is also effective. However, traditional techniques may alter colors in regions that are not affected by smoke.

Deep learning, particularly GANs, offers improved results. Goodfellow et al. [14] introduced GANs to address blurring in CNN outputs, achieving photo-realistic quality. Techniques like pix2pix [9] and CycleGAN incorporate perceptual and cycle loss to enhance de-smoking performance. This framework has potential for smoke removal, with hazy and clear data serving as paired inputs.

Recent advancements in laparoscopic video quality include LVQIS by Zheng et al. [15], which uses GANs for motion blur and smoke removal, and the approach by Li et al. [16], combining contrast enhancement with multi-level frequency learning to improve real-time smoke removal in surgery. Xia et al. [17] introduced a dataset of smoky/smoke-free scenes to benchmark de-smoking methods, while Wu et al. [18] developed SelfSVD, a self-supervised technique leveraging pre-smoke frames, achieving effective smoke removal and detail restoration.

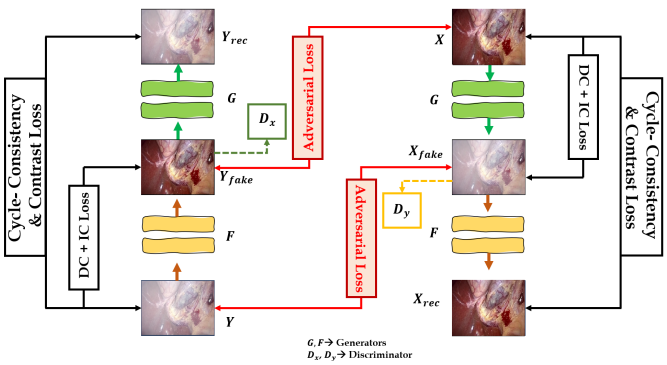

Fig. 1. An overview of the proposed method, highlighting how inter-channel (IC) differences and the dark channel (DC) prior are utilized to assess the residual smoke, supporting cycle-consistency and adversarial losses for effective smoke removal.

III. Methodology

The proposed model leverages the CycleGAN architecture for unpaired image-to-image translation between two domains, incorporating additional loss functions to enhance the desmoking process. Specifically, inter-channel difference and dark channel prior losses are introduced to further refine the generated images by targeting residual smoke artifacts and promoting optimal image generation (refer to Fig. 1)

A. Cycle-consistent generative adversarial network

Our method is based on the CycleGAN architecture, which facilitates unpaired image-to-image translation between source domain $\mathcal{A}$ and target domain $\mathcal{B}$ using adversarial and cycle-consistency losses. In CycleGAN, the generator creates realistic images to deceive the discriminator, while the discriminator distinguishes between real and generated images, updating their parameters to maintain equilibrium. Given unpaired clear images $\{a_i\}_{i=1}^{N}$ and hazy images $\{b_j\}_{j=1}^{M}$, the goal is to map $\mathcal{A}$ to $\mathcal{B}$. Discriminators $D_{\mathcal{A}}$ and $D_{\mathcal{B}}$ differentiate between real images and those generated from the opposite domain. The adversarial loss, applied to both generator and discriminator, measures the difference between generated and real images:

$$\begin{aligned}\mathcal{L}(G, D_{\mathcal{B}}, \mathcal{A}, \mathcal{B}) &= \mathbb{E}_{b\sim p_{\text{data}}(b)}[\log D_{\mathcal{B}}(b)] \\ &\quad + \mathbb{E}_{a\sim p_{\text{data}}(a)}[\log(1 - D_{\mathcal{B}}(G(a)))], \quad (2)\\ \mathcal{L}(F, D_{\mathcal{A}}, \mathcal{A}, \mathcal{B}) &= \mathbb{E}_{a\sim p_{\text{data}}(a)}[\log D_{\mathcal{A}}(a)] \\ &\quad + \mathbb{E}_{b\sim p_{\text{data}}(b)}[\log(1 - D_{\mathcal{A}}(F(\mathcal{B})))], \quad (3)\end{aligned}$$

where $a \sim p_{\text{data}}(a)$ and $b \sim p_{\text{data}}(b)$ represent the data distributions of the source and target domains, respectively. The generative adversarial loss objective is:

$$\begin{aligned}\mathcal{L}_{\text{GAN}}(G, F, D_{\mathcal{A}}, D_{\mathcal{B}}) &= \min_{G} \max_{D_{\mathcal{B}}} \mathcal{L}(G, D_{\mathcal{B}}, \mathcal{A}, \mathcal{B}) \\ &\quad + \min_{F} \max_{D_{\mathcal{A}}} \mathcal{L}(F, D_{\mathcal{A}}, \mathcal{B}, \mathcal{A}). \quad (4)\end{aligned}$$

Cycle-consistency loss evaluates if data in one domain, after forward and reverse translation, approximates the original. It ensures recovery through the forward $a \to G(a) \to F(G(a)) \approx a$ and backward $b \to F(b) \to G(F(b)) \approx b$ cycles:

$$\begin{aligned}\mathcal{L}_{\text{cyc}}(G, F) &= \mathbb{E}_{a\sim p_{\text{data}}(a)}[\|a - F(G(a))\|_1] \\ &\quad + \mathbb{E}_{b\sim p_{\text{data}}(b)}[\|b - G(F(b))\|_1]. \quad (5)\end{aligned}$$

B. Desmoking in laparoscopic surgery

CycleGAN alone struggles with smoke removal in laparoscopic videos because it lacks specific smoke-related priors. To address this, we propose a desmoking approach that enhances CycleGAN with two additional loss functions: inter-channel loss (IC) and dark-channel loss (DC). These losses assess remaining smoke components by evaluating pixel differences pre- and post-processing.

1) Inter-channel (IC) Loss: Inter-channel discrepancies are defined as the absolute difference between color channels:

$$\Delta(C) = \|C_{\text{red}} - C_{\text{green}}\|_1 + \|C_{\text{green}} - C_{\text{blue}}\|_1 + \|C_{\text{blue}} - C_{\text{red}}\|_1, \quad (6)$$

where C represents a pixel and $C_{\text{red}}, C_{\text{green}}, C_{\text{blue}}$ are its RGB channel values, normalized between 0 and 1. The average inter-channel loss for an image A is:

$$L^{IC}(A) = \frac{1}{m}\sum_{j=1}^{m} \Delta(C_j), \quad (7)$$

where C_j represents individual pixels in the image, and m is the total number of pixels. Low IC values indicate higher smoke levels, prompting the generator to minimize smoke during hazy-to-clear transformation. The IC loss in our model is:

$$\mathcal{L}_{IC} = g\left(L^{IC}(G(a))\right) + g\left(L^{IC}(F(b))\right), \quad (8)$$

where

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$$g(x) = \begin{cases} 1 - \|x-1\|, & \text{clear} \rightarrow \text{hazy}, \\ \|x-1\|, & \text{hazy} \rightarrow \text{clear}. \end{cases} \tag{9}$$

2) Dark-channel (DC) Loss: Inspired by the dark channel prior, DC loss measures hazy components by identifying the minimum intensity across RGB channels:

$$I^{\min}(p) = \min_{q \in \mathcal{N}(p)} \left(\min_{c \in \{r,g,b\}} I^c(q) \right), \tag{10}$$

where I^c denotes the intensity of each color channel within a local neighborhood $\mathcal{N}(p)$. For clear images, $I^{\min} \rightarrow 0$. The DC loss, applied to discriminators, becomes large when smoke dominates the image, guiding the generator to minimize smoke:

$$\mathcal{L}_{\text{min}} = I^{\min}(F(b)) + I^{\min}(G(a)). \tag{11}$$

This enhanced CycleGAN model uses both IC and DC losses to improve smoke-free translation in laparoscopic video data.

3) Combined Loss Function: The overall loss function for the proposed Desmoke-LAP is expressed as:

$$\begin{aligned} \mathcal{L}(G', F', D'_X, D'_Y) = \mathcal{L}_{\text{CycleGAN}}(G', F', D'_X, D'_Y) \\ + \alpha' \cdot \mathcal{L}_{IC'}(D'_X, D'_Y) \\ + \beta' \cdot \mathcal{L}_{DC'}(D'_X, D'_Y), \end{aligned} \tag{12}$$

where

$$\mathcal{L}_{\text{CycleGAN}}(G', F', D'_X, D'_Y) = \mathcal{L}_{\text{GAN}}(G', F', D'_X, D'_Y) + \mathcal{L}_{\text{cyc}}(G', F'). \tag{13}$$

The parameters α' and β' control the balance between IC', cycle, and DC' losses, with β' tuned experimentally. Here, G' and F' represent the generators for domains X' and Y', while D'_X and D'_Y are the discriminators in each domain.

C. Implementation Details

The proposed approach was executed utilizing PyTorch version 0.4. During the training phase, computations were accelerated by an 11 GB GeForce RTX 2080 Ti GPU. The network was trained for 50 epochs using the ADAM optimizer, with a learning rate set to 0.0002 and a momentum parameter of 0.5. A consistent batch size of 4 was applied during the entire training process to ensure stability. To accommodate the input dimensions of 256 by 256 pixels, the original images were resized accordingly and supplemented with zero-padding.

D. Dataset

This study utilizes the LVQ database, derived from the Cholec80 dataset, featuring a selection of videos that encompass a range of laparoscopic scenes and actions, including burning, organ extraction, irrigation, and cutting, among others. The LVQ database includes 10 reference videos (each 10 seconds long) and five types of distortions: motion blur, defocus, smoke, noise, and lighting inconsistency, applied at four intensity levels. Videos are stored uncompressed in .avi format at 512x288 resolution, using screen blending to simulate smoke distortions by overlaying dark backgrounds. The study uses the Saturation Analysis (SAN) classifier to detect smoke in frames by analyzing histogram saturation. The smoke level in a frame is assessed with the threshold 0.35, yielding probabilities $p(S)$ and $p(NS)$ for smoke presence or absence:

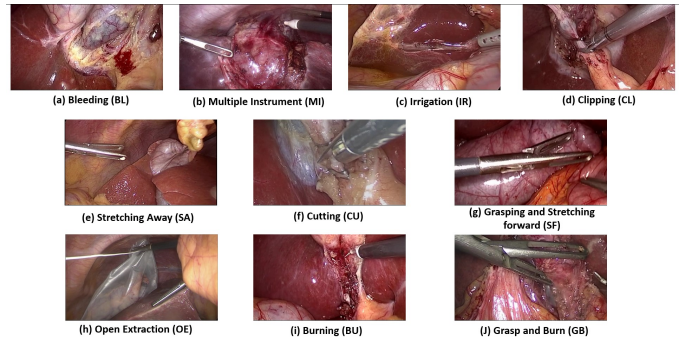

Fig. 2. Reference images of each frame samples in the LVQ database [8], [19]

TABLE I
COMPARATIVE ANALYSIS USING IMAGE QUALITY METRICS.

Ref	PSNR	SSIM	JNBM	FADE	REA
[2]	26.42	0.65	0.32	0.48	0.48
[7]	19.87	0.67	0.40	0.49	0.48
[9]	21.3	0.78	0.48	0.58	0.57
[20]	24.8	0.82	0.34	0.54	0.58
Ours	**38.6**	**0.94**	**0.75**	**0.26**	**0.75**

$$p(S) = \frac{1}{|hist|} \sum_{\substack{i=0 \\ b \in hist, t \leq t_c}} b_i \tag{14}$$

$$p(NS) = 1 - p(S) \tag{15}$$

Synthetic smoke datasets generated from LVQ were used, with reference frames as ground truth and varied smoke-distorted images for training. From 5122 frames, 70% were for training, and 30% were used for testing.

IV. EXPERIMENTAL RESULT

This section presents both qualitative and quantitative evaluations of the desmoked images using three referenceless image quality metrics: FADE, REA [20], and JNBM. FADE evaluates fog density, with higher values indicating greater fog. JNBM measures perceived sharpness, with lower values indicating less blurriness, while REA assesses edge restoration, with higher values indicating better restoration. Additionally, PSNR and SSIM are used to compare the results with reference videos. Optimal performance is characterized by lower FADE values and higher scores in PSNR, JNBM, SSIM, and REA. Fig. 3 shows that the proposed method outperforms previous techniques, significantly improving visibility in thick smoke scenarios.

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Due to the significant time constraints faced by surgeons, we optimized the recruitment process and enhanced participation by employing a concise online survey. This web-based format allowed for broader involvement in data collection and feedback. Invitations were sent to 25 surgeons, all of whom consented to participate. The survey was structured into two sections, each containing ten image comparison questions. For each question (Fig. 4), surgeons were asked to select the image they (a) preferred and (b) considered most suitable for surgical applications. The results revealed that our proposed method received the most favorable responses in both categories, underscoring its effectiveness.

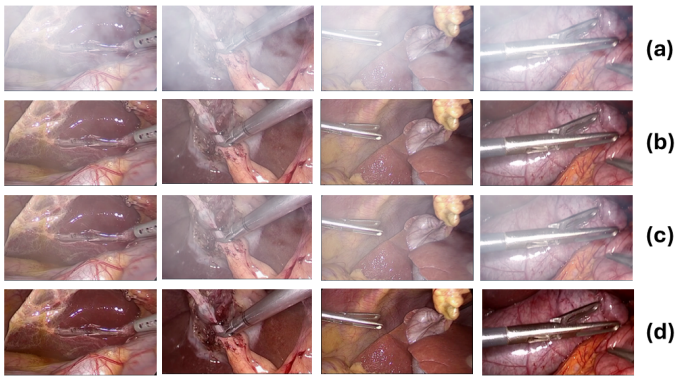

Fig. 3. Comparison of the suggested approach utilizing the LVQ database with current models. Different frames from left to right of the LVQ dataset and (a) Input, (b) [7], (c) [20], (d) Our method

V. Conclusion

In this work, we present a GAN-based method for real-time smoke elimination in laparoscopic procedures, tackling the visual obstructions from electrocauterization. Our approach uses a CycleGAN framework with inter-channel and dark channel prior losses to minimize residual smoke, without relying on synthetic paired data or atmospheric models. Comparative analysis shows our model produces smooth, consistent desmoked frames, outperforming existing methods. A user study with surgeons validated the method's clinical potential, highlighting its ability to deliver high-quality, smoke-free video outputs that enhance surgical safety and efficiency. The dataset and code will be made publicly available for future research.

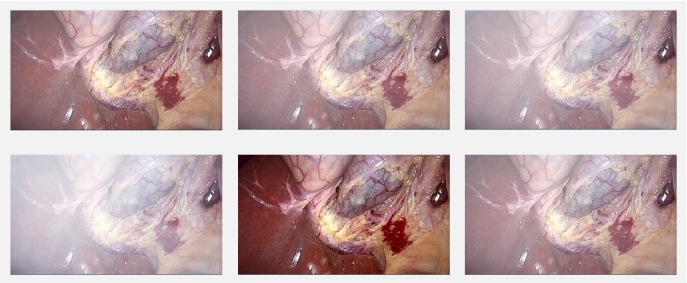

Fig. 4. Visual representation of the online experiment platform [1]

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